

GLOBAL QUANTUM MECHANICS CHALLENGE 2026

Experimental Study of Quantum Error Correction Under Qubit Noise

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Abstract

This project investigates whether a simple quantum error-correction method can reduce the effect of bit-flip noise. A three-qubit repetition code encodes one logical qubit across three physical qubits. A Monte Carlo simulation was performed with 100,000 trials for each physical error probability. The simulated logical error rate was compared with an unencoded qubit and with the theoretical expression $P_L = 3p^2 - 2p^3$. Statistical uncertainty was estimated using a binomial standard error and 95% confidence intervals. The simulation demonstrates the expected reduction in logical errors at low physical error rates, while also showing the limitation of the simple code at higher noise levels. The numerical results in this report are simulated data, not measurements from a physical quantum processor.

1. Research Question

How does a three-qubit quantum bit-flip error-correction code affect logical error probability as the physical bit-flip error rate increases?

2. Hypothesis

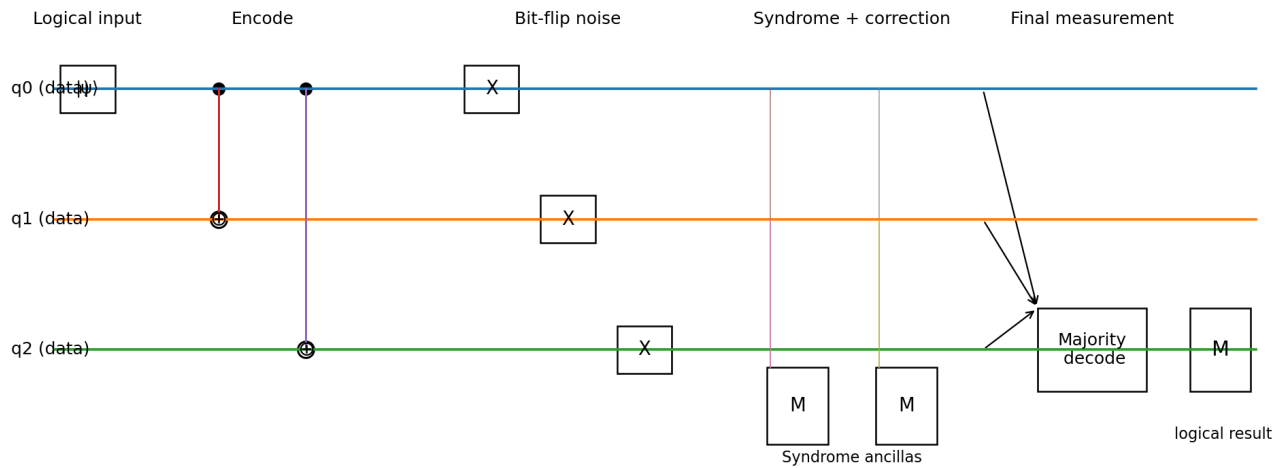
For sufficiently small independent physical bit-flip probability p , majority-based correction using three physical qubits will produce a logical error probability lower than p .

3. Objectives

Construct an unencoded reference model; encode one logical qubit using a three-qubit repetition code; apply controlled bit-flip noise; decode by majority; compare simulated logical error rates with theory; quantify statistical uncertainty; and discuss the limitations of the code.

4. Theory

A qubit can be written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. A bit-flip is represented by the Pauli-X operation, $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. The three-qubit code maps $|0_L\rangle = |000\rangle$ and $|1_L\rangle = |111\rangle$, giving $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$. This is an encoding, not cloning an arbitrary unknown state.



6. Simulation Method

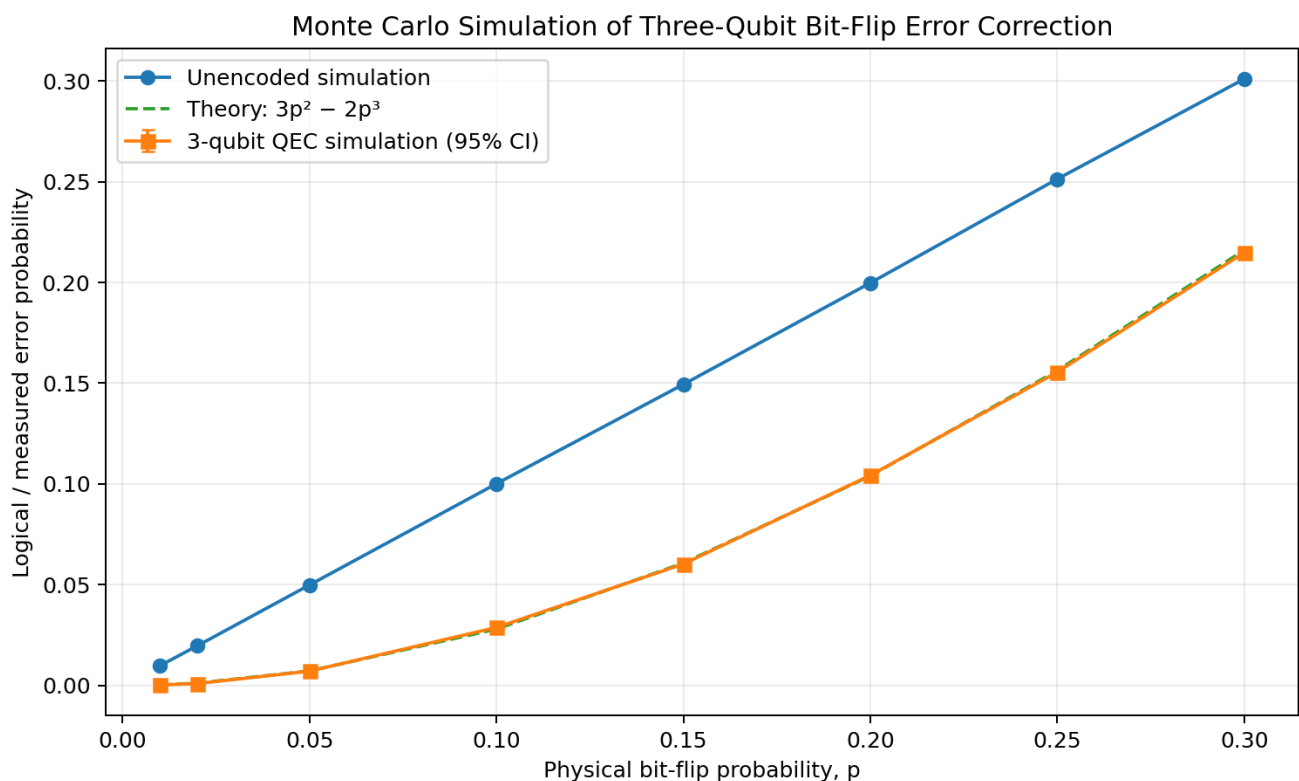
A Monte Carlo simulation was performed using a fixed random seed for reproducibility. For each physical error probability $p \in 0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30$, 100,000 independent trials were generated. In the unencoded model, one qubit was subjected to a bit flip with probability p . In the encoded model, three physical qubits independently experienced bit flips with probability p ; majority decoding was then applied. A logical error was counted when at least two of the three physical qubits flipped. This simulation directly implements the independent bit-flip noise model.

7. Theoretical Prediction

For the unencoded qubit, $P_{\text{uncorrected}}=p$. For the three-qubit code, failure occurs when two or three physical qubits flip: $P_L=3p^2(1-p)+p^3=3p^2-2p^3$.

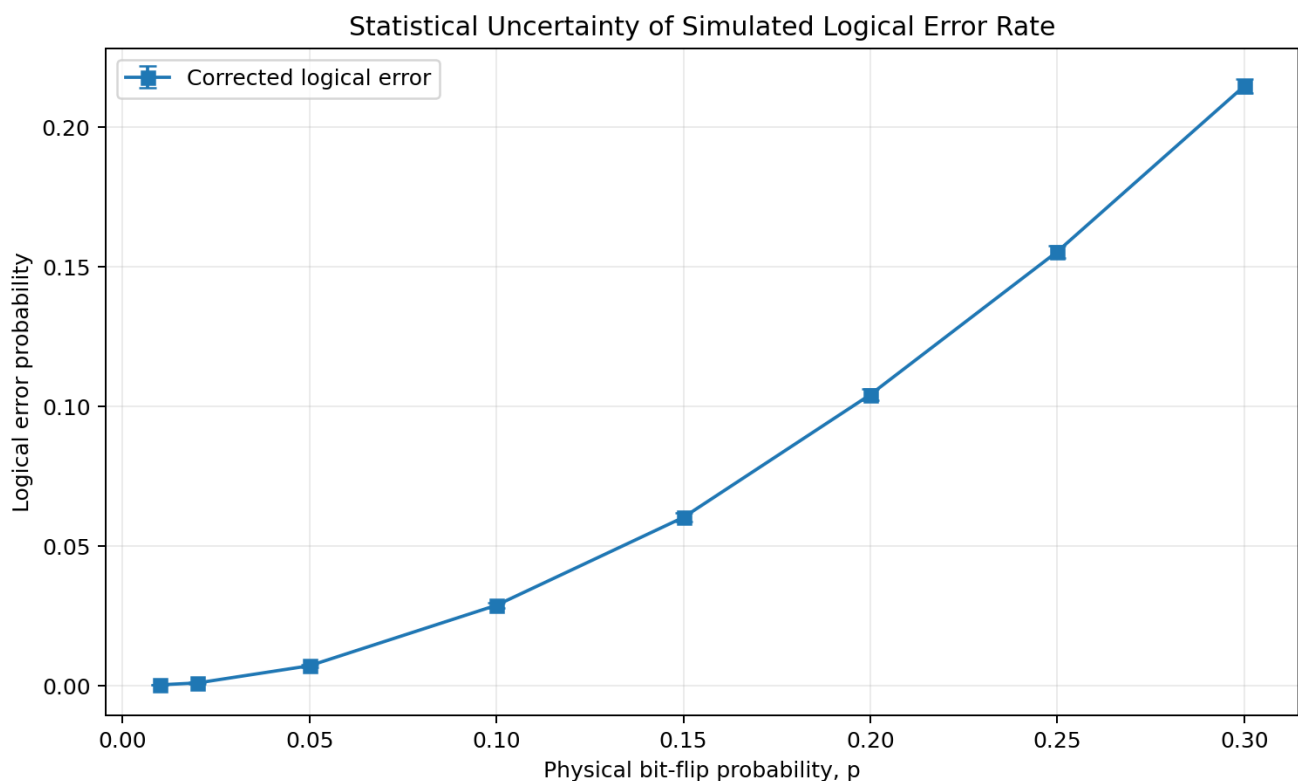
8. Simulated Results

The graph below compares the simulated unencoded error rate, simulated corrected logical error rate, and the theoretical prediction. The corrected simulation follows the theoretical curve within expected Monte Carlo fluctuations.



Simulation data and uncertainty table

p	Unenc.	SE	Corrected	SE	Theory
0.01	0.00987	0.00031	0.00032	0.00006	0.00030
0.02	0.01973	0.00044	0.00099	0.00010	0.00118
0.05	0.04990	0.00069	0.00715	0.00027	0.00725
0.10	0.10014	0.00095	0.02883	0.00053	0.02800
0.15	0.14944	0.00113	0.06031	0.00075	0.06075
0.20	0.19978	0.00126	0.10429	0.00097	0.10400
0.25	0.25131	0.00137	0.15552	0.00115	0.15625
0.30	0.30094	0.00145	0.21483	0.00130	0.21600



10. Interpretation

At low physical error rates, the three-qubit code reduces logical errors because a single isolated bit flip can be corrected. As p increases, two-or-more-error events become more common and the improvement decreases.

11. Limitations

The simulation uses independent bit-flip noise and idealized correction. It does not represent all hardware effects or arbitrary quantum errors. The three-qubit repetition code is therefore an educational demonstration, not a complete fault-tolerant architecture.

12. Connection to Modern Quantum Computing

IBM Quantum documents the repetition code as a simple QEC example, while modern surface-code experiments investigate logical-error suppression using many physical qubits. The present study demonstrates the same core principle at small scale.

13. Conclusion

The Monte Carlo simulation supports the hypothesis under the specified independent bit-flip noise model. The corrected logical error rate is lower than the unencoded error rate at low physical error probabilities, while the model also illustrates the limitations of simple error correction.

14. Future Work

Investigate phase-flip and depolarizing noise, noisy syndrome measurements, larger codes, the Shor code, and surface-code architectures or real quantum hardware.

15. References

IBM Quantum Documentation, "Repetition codes."

IBM Quantum Learning, "Repetition codes."

Google Quantum AI, "Suppressing quantum errors by scaling a surface code logical qubit," *Nature* 614, 676–681 (2023).

Google Quantum AI and collaborators, "Quantum error correction below the surface code threshold," *Nature* 638, 920–926 (2025).